VLSI Design



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About the Program

Welcome to Nyeras – your passport to VLSI Design expertise!

At Nyeras, we're on a mission to equip you with the skills needed to thrive in the ever-evolving world of VLSI Design. Our courses are crafted by industry experts who know the ins and outs of digital defense.

Get ready for hands-on learning that brings VLSI Design to life. Whether you're a newbie or a seasoned pro, our flexible learning options fit your schedule and style.

Join us and become a Tech guardian. Let's secure the digital frontier together!

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key Features

- Exclusive Access to LMS Portal
- Live Interactive Classes
- 7+ Years Industrial Expert Trainers
- Dedicated Course Materials
- 2 - Realtime Projects with Internship
- Mentorship till Placement

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The Roadmap



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Course Curriculum

1 Introduction to VLSI

- Overview of VLSI design flow and its importance
- Evolution of VLSI technology and its applications.

2 Digital Design Fundamentals

- Number systems and codes.
- Boolean algebra and logic gates.
- Combinational and sequential circuits.
- Karnaugh maps and simplification techniques.

Course Curriculum

3 Basic Electronics

- Semiconductor basics: doping, PN junction, MOS transistors.
- MOSFET operation and characteristics.
- CMOS (Complementary Metal-Oxide-Semiconductor) technology.

4 Circuit Analysis

- Basic circuit analysis techniques.
- RC and RL circuits.
- Transient and steady-state analysis.

Course Curriculum

5 Basic CMOS Logic Design

- CMOS inverter and its characteristics.
- CMOS logic gates: NAND, NOR, AND, OR.
- Layout design principles.

6 VLSI Design Methodologies

- RTL (Register Transfer Level) design.
- Behavioral, structural, and physical design.
- Design hierarchy and abstraction levels.

Course Curriculum

7 Digital System Design

- Finite State Machines (FSMs).
- Datapath and control unit design.
- Memory design: SRAM, DRAM

8 Timing Analysis and Optimization

- Static timing analysis (STA).
- Clock domain crossing (CDC) analysis.
- Timing closure techniques.

Course Curriculum

9 ASIC Design Flow

- RTL synthesis.
- Technology mapping and optimization.
- Physical design: floorplanning, placement, routing

10 Verification Techniques

- Functional verification.
- Simulation and testing.
- Formal verification methods.

Course Curriculum

11 Low Power Design Techniques

- Power dissipation sources.
- Clock gating, voltage scaling.
- Power management strategies

12 High-Level Synthesis (HLS)

- C-based design and synthesis.
- Trade-offs between HLS and RTL design.

Course Curriculum

13 Analog and Mixed-Signal IC Design

- Basics of analog circuit design.
- Mixed-signal integration challenges.
- Behavioral modeling and simulation.

14 Advanced Topics in VLSI

- Nanotechnology and emerging devices.
- 3D IC integration.
- Quantum computing and its implications for VLSI

Get in touch
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